Xuanyu **Lyu**

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Education

University of Colorado-Boulder | Institute for Behavior Genetics — Start from Aug. 2023

Ph.D. in Behavioral, Psychiatric, and Statistical Genetics, Expected May. 2028

- Advisor: Dr. Matthew C. Keller
- Research Area: Developing Novel Computational and Statistical Methods in Complex Trait Modeling

University of Colorado-Boulder — Start from Aug. 2024

MS in Applied Mathematics, Expected May. 2027

- Research Area: Long-term Prediction of Anxiety Using Wearable Device Data: Development of New Machine Learning Methods

Wake Forest University — Aug. 2021 - Aug. 2023

M.A. in Psychology

- Advisor: Dr. S. Mason Garrison
- Research Area: Statistical Methods and Behavior Genetics
- Master's Thesis:
  Family structures impact statistical power of mitochondrial DNA effect estimation under a novel extended-pedigree model

Shandong University — Sep. 2016 - June. 2020

B.A. in Sociology

- First-Year Major: Software Engineering
- Undergraduate Thesis (Awarded Shandong University 2020 Outstanding Undergraduate Thesis):
  The Relationship between Social Media, Pornography, and Self-Objectification-An investigation based on structural equation modeling

The University of Queensland — Jan. 2019 - July. 2019

Study Abroad

- Courses: Social Psychology, Attitudes and Social Cognition, Learning and Cognition, Psychological Methodology III

Research Interests

Methods	Deep Learning; Simulation-Based Inference; Statistical & Computational Genetics; Family-Based and Genetically-Informed Designs
Applications	Polygenic Scores; Pharmacogenomics & Precision Medicine; Indirect Genetic Effects & Assortative Mating; Mitochondrial DNA Effects

Publication and Manuscripts

- **Lyu, X.**, Balbona, J., Chen, T. & Keller, M. C. *eLife* Multivariate SEM-PGS Model: Using Polygenic Scores to Investigate Cross-Trait Indirect Genetic Effects. doi: 10.7554/eLife.109470.1
- Hunter, M. D., Garrison, S. M., **Lyu, X.**, Good, R., Mylakumar, M. N., Moyakhe, L. B., Carroll, S. L. & Burt, S. A. (2026). Tracing the Right Path: Determination of Large Pedigree Segmentation and Relatedness. *Behavior Genetics*. doi: 10.1007/s10519-026-10259-z
- Aczel, B., Szaszi, B., …, **Lyu, X.**, …& Nosek B.A. (2026). Investigating the analytical robustness of the social and behavioural sciences. *Nature*, 652(8108), 135–142. doi: 10.1038/s41586-025-09844-9
- Chen, T., **Lyu, X.** & Keller, M. C. (First Draft Finished.) Comparing Forward-Time and Model-Implied Covariance-Based Simulations for Evaluating Family-Based Genetic Models
- **Lyu, X.**, Hunter, M. D., Burt, S. A., Good, R., Carroll, S. L., & Garrison, S. M. (2025) Detecting mtDNA effects with an Extended Pedigree Model: An Analysis of Statistical Power and Estimation Bias. *Behavior Genetics.* doi:10.1007/s10519-025-10225-1
- Burt, S. A., Garrison, S. M., **Lyu, X.**, Rodgers, J. L., Carroll, S. L., Smith, Ken R., & Hunter, M. D. (2025) Contributions of inherited mtDNA to longevity: evidence from extended pedigrees with 176 million kinship pairs. *EBioMedicine.* doi:10.1016/j.ebiom.2025.105911
- Garrison, S. M., Hunter, M. D., **Lyu, X**., Trattner, J. D., & Burt, S. A. (2024). BGmisc: An R Package for Extended Behavior Genetics Analysis. *Journal of Open Source Software.* doi: 10.21105/joss.06203

- Garrison, S.M., Robertson, H., **Lyu, X.**, Trattner, J., McKinzie, L., & Rodgers, J.L.. (2024). Using genetically-informed designs to test causal claims without experiments: Discordant-Sibling designs and applications. Revise and resubmit to *AMPSS.* doi: 10.1101/2025.08.25.25334395
- **Lyu, X.** (2023). Pedigree Structures Impact Statistical Power of MtDNA Effects Estimation *(Master's Thesis, Wake Forest University).* Available from ProQuest Dissertations & Theses A&I
- **Lyu, X.** & Garrison, S.M.(2023). Effects of Genetic Relatedness of Kin Pairs on Univariate ACE Model Performance. *Twin Research and Human Genetics, 26(4-5)*, 257-268. doi:10.1017/thg.2023.40

Research Talks

Lyu, X., Chen, T., Smajlagic, D., Valenza-Troubat, N., & Keller, C.M., A Statistical Review of Indirect Effects Models with Simulation — June. 2025
Talk Given at a Symposium at the 55th Behavior Genetics Association Annual Meeting · Atlanta, USA

Lyu, X., Chen, T., Smajlagic, D., Valenza-Troubat, N., & Keller, C.M., Advancements in Multivariate SEM-PGS Model: New Insights into Familial Resemblance — June. 2025
Talk Given at a Symposium at the 55th Behavior Genetics Association Annual Meeting · Atlanta, USA

Eftedal, N.H., **Lyu, X.**, & Keller, C.M., Using Extended Twin Family: What correlations between relatives-in-law on polygenic scores can tell us about assortative mating — June. 2025
Talk Given at a Symposium at the 55th Behavior Genetics Association Annual Meeting · Atlanta, USA

Chen, T., **Lyu, X.**, & Keller, C.M., Using Extended Twin Family Designs to Differentiate Multiple Causes of Parent-Offspring Similarity — June. 2025
Talk Given at a Symposium at the 55th Behavior Genetics Association Annual Meeting · Atlanta, USA

Garrison, S.M., Mylakumar, N., **Lyu, X.**, Gatz, M., Hunter, M., S. Burt, S.A., Tracing Mitochondrial Inheritance of Alzheimer's Disease: A Family-Based Case-Control Study of 4.8 Million Cousins — June. 2025
Talk Given at a Symposium at the 55th Behavior Genetics Association Annual Meeting · Atlanta, USA

Lyu, X., Symposium Title: Beyond Nuclear Families: What Do We Gain from Extended Family Members in Behavioral Genetic Research? — June. 2025
Chair of a Symposium at the 55th Behavior Genetics Association Annual Meeting · Atlanta, USA

Lyu, X., Balbona, J., Kim, Y. & Keller, C.M., Multivariate SEM-PGS Model: Using Polygenic Scores to Demystify Cross-Trait Genetic Nurture — June. 2024
Talk Given at a Symposium at the 54th Behavior Genetics Association Annual Meeting · London, the United Kingdom

Lyu, X. & Noemie Valenza-Troubat. GitHub Tutorial Workshop at IBG — Nov. 2023
Presentation Given at StatGen Meeting at IBG · Boulder, US

Lyu, X. Genetic Nurture: New Insights from Family Design in Post-GWAS Era — Nov. 2023
Guest Lecture in PSYC 3102-Behavior Genetics Course · Boulder, US

Lyu, X. & Garrison, S.M. Effects of Genetic Relatedness of Kin Pairs on Univariate ACE Model Performance. — Sep. 2023
Talk Given at the Equinox Conference 2023 · Online

Garrison, S.M., **Lyu, X.**, Hunter, M.D., Rodgers, J.L, Smith, K.R., Coon, H., & Burt, S.A. Analyzing extended cousin similarity to unravel the mystery of mtDNA and longevity. In Burt, S.A. (Chair), A new extended pedigree model for quantifying the contributions of mitochondrial DNA. — June. 2023
Talk Given at Symposium at the 53rd Behavior Genetics Association Annual Meeting · Murcia, Spain

Hunter, M.D., **Lyu, X.**, Garrison, S.M., Rodgers, J.L, Smith, K.R., Coon, H., & Burt, S.A. Modeling mtDNA Effects from Extended Pedigrees in the Utah Population Database. In Burt, S.A. (Chair), A new extended pedigree model for quantifying the contributions of mitochondrial DNA. — June. 2023
Talk Given at Symposium at the 53rd Behavior Genetics Association Annual Meeting · Murcia, Spain

Lyu, X., Hunter, M.D., Rodgers, J.L, Smith, K.R., Coon, H., Burt, S.A., & Garrison, S.M. Statistical power analysis on mtDNA effects estimation. In Burt, S.A. (Chair), A new extended pedigree model for quantifying the contributions of mitochondrial DNA. — June. 2023
Talk Given at Symposium at the 53rd Behavior Genetics Association Annual Meeting · Murcia, Spain

Lyu, X. & Garrison, S.M. Leveraging the China Family Panel Study: An Estimation of Height Heritability Using Preliminary Kinship Links — June. 2022

Talk Given at the Quantitative Methods Section of Behavior Genetics Association 2022 Annual Meeting · Los Angeles, US

Lyu, X. Social Media Use Predicts Increased Self-objectification through Social Comparison. — Oct. 2019

Oral Presentation Earned the 2nd prize of Outstanding Paper at 11th Chinese Psychologist Conference · Nanjing, China

Research Posters

Lyu, X., Eftedal, N.H., Huang, F., & Keller, C.M. Neural Posterior Estimation for Complex Trait Modeling: Resolving Excessive Educational Attainment Polygenic Score Correlations in Extended Kinship — Oct. 2026

Poster Presented at American Society of Human Genetics 2026 Annual Meeting · Montreal, Canada

Lyu, X., Eftedal, N.H., Huang, F., & Keller, C.M. Deciphering Kinship Polygenic Score Correlations: A Deep Learning and Agent-based Framework for Behavior Genetics — May. 2026

Poster Presented at ASA-STATGEN 2026 Annual Meeting · Montreal, Canada

Lyu, X., Chen, T. & Keller, C.M. Statistical Review on Models Quantifying Indirect Genetic Effects - A Comprehensive Simulation Study — Oct. 2025

Poster Presented at American Society of Human Genetics 2025 Annual Meeting · Boston, US

Lyu, X., Balbona, J., Chen, T., Kim, Y. & Keller, C.M. Multivariate SEM-PGS Model: Using Polygenic Scores to Investigate Cross-Trait Indirect Genetic Effects — Nov. 2024

Poster Presented at American Society of Human Genetics 2024 Annual Meeting · Denver, US

McKinzie, L, **Lyu, X.**& Garrison, S.M. Leveraging the power of sibling data to untangle the relationship between ADHD, SES, and academic achievement — June. 2022

Online Poster Presented at Behavior Genetics Association 2022 Annual Meeting · Los Angeles, US

Milhollen, M., **Lyu, X.** & Garrison, S.M. The China Family Panel Study: An Opportunity to Combat WEIRDNESS in Behavior Genetics — June. 2022

Online Poster Presented at Behavior Genetics Association 2022 Annual Meeting · Los Angeles, US

Lyu, X. & Garrison, S.M. Leveraging the China Family Panel Study: An Estimation of Height Heritability Using Preliminary Kinship Links — April. 2022

First-Year Project Poster Session for Psychology MA at Wake Forest University · Winston Salem, US

Lyu, X. The process towards self-objectification: The role of social comparison and body surveillance in the relationship between social media use and body shame. — Feb. 2020

Poster Accepted at Society of Personality and Social Psychology 2020 Annual Convention · New Orleans, US

Software

- Garrison, S. M., Hunter, M. D., **Lyu, X.**, Trattner, J. D., & Burt, S. A. (2023). BGmisc: An R Package for Extended Behavior Genetics Analysis (1.0.1). [Computer software] https://cran.r-project.org/web/packages/BGmisc/index.html
- **Lyu, X.** & Garrison, S.M.(2022). ACEsimFit: ACE Kin Pair Data Simulations and Model Fitting. [Computer software] https://CRAN.R-project.org/package=ACEsimFit

Teaching Experience

University of Colorado at Boulder

Teaching Assistant at Psychology Department — August. 2023 - Dec. 2023

- Teaching Assistant for PSYC 3102-Behavioral Genetics

Wake Forest University

Teaching Assistant at Psychology Department — August. 2021 - May. 2022

- Courses: Methods in Psychological Research (2021 Fall; 2022 Spring), Personality (2021 Fall), Research in Social Psychology (2022 Spring), Mass Testing (2021 Fall; 2022 Spring)

Working Experience

Merck Rahway, US

Biostatistics Intern at Pharmacogenetics Group May. 2026 - August. 2026

- Develop pharmacogenetics methods to construct sparse PRS for drug response prediction in clinical settings

Tencent Shanghai, China

UX Researcher at Interactive Entertainment Group Jan. 2021 - July. 2021

- Conducted user experience surveys, extracted user back-end data, analysed data integrally and create data analysis report

Kwai Beijing, China

UX Research Intern at Data Analysis Department Sep. 2020 - Jan. 2021

- Optimized the data cleaning/data analysis process using R, and wrote reusable data cleaning/weighting scripts to increase the efficiency of the data processing process by over 50% and improve the accuracy of the data processing workflow.

Beijing Normal University Beijing, China

Research Assistant at Laboratory of Cognitive Neuroscience and Learning June. 2020 - Sept. 2020

- Used R, Python and other tools to realize data collection, data extraction, data cleaning, data analysis, data visualization for various projects in the lab.

Honors & Awards

June. 2026	2025 Editors' Choice Award for a Paper Published in Behavior Genetics ($1000)	*Award*
Sept. 2023	2023 Diversity Recruitment Fellowship from the Psychology & Neuroscience Department of CU-Boulder ($3750)	*Scholarship*
June. 2023	2023 Behavioral Genetics Annual Conference Travel Award ($350)	*Award*
June. 2022	Wake Forest University Research Assistantship Scholarship ($50871)	*Scholarship*
August. 2021	Wake Forest University Teaching Assistantship Scholarship ($50573)	*Scholarship*
July. 2020	Shandong University 2020 Outstanding Undergraduate Thesis	*1st Class*
Oct. 2019	Outstanding Paper of 11th Chinese Psychologist Conference	*2nd Prize*
Oct. 2019	Overseas Study Scholarship of Shandong University (≈$450)	*1st Class*
Nov. 2018	The 4th China College Students' Internet-Plus Innovation and Entrepreneurship Competition	*2nd Prize*
Oct. 2018	Scholarship for Outstanding Students (≈$300)	*2nd Class*
Oct. 2017	Scholarship for Outstanding Students (≈$300)	*2nd Class*
May. 2017	2017 Enactus Entrepreneurial Action & Social Innovation Competition (China Region)	*2nd Prize*

Skills

Statistical Methods	Neural posterior estimation; Bayesian computation; Penalized regression & ensemble methods; Causal inference (SCMs, Bayesian networks); Linear mixed-effects models; Simulation studies & power analysis; Structural equation modeling; Genetically informed designs
Programming	Python (PyTorch, sbi, NumPy, pandas, SciPy), R (OpenMx), MATLAB
Computing	Linux/Unix, Slurm/HPC, Git & GitHub, LLM coding agents for research workflows
Genomics Tools	PLINK, GCTA, LDAK, REGENIE, MAGMA, LDpred2, Ensembl VEP, BLAST, AlphaFold